

Requirements Table

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Red ID	TYPE	Description	priority	acceptance criteria	rationale
FR-001	Functional	The system shall ingest inbound webhooks, forward payloads to target endpoints, and automatically retry failed deliveries using exponential backoff (1s, 2s, 4s, 8s).	high	Pass: Failed delivery successfully retried and logged; Fail: Dropped webhook without dead-letter persistence.	Core reliability guarantee of the gateway; prevents silent event loss.
FR-002	Functional	The system shall route failed webhooks to a Dead-Letter Queue (DLQ) after all exponential backoff retries are exhausted.	high	Pass: Undeliverable webhooks appear in the DLQ with failure metadata after 4 retries. Fail: Payload is lost or discarded without reaching the DLQ.	Prevents data loss for permanently failing endpoints while keeping main processing queues clear.
FR-003	Functional	The system shall log all incoming request payloads, delivery attempts, HTTP status codes, and execution timestamps.	medium	Pass: Audit logs correctly capture request payload and response statuses for 100% of incoming calls. Fail: Logs are missing, truncated, or untimed.	Necessary for debugging integration failures and providing system transparency to Integration Developers.
FR-004	Functional	The system shall allow System Operators to manually replay webhooks stored in the Dead-Letter Queue (DLQ).	medium	Pass: Clicking 'Replay' re-enqueues the selected payload and triggers target delivery. Fail: Replay fails or alters the original payload content.	Enables manual recovery after downstream integration endpoints recover from prolonged outages.
FR-005	Functional	The system shall authenticate incoming webhook requests using signature verification (e.g., HMAC-SHA256 headers).	high	Pass: Invalid or missing signatures yield HTTP 401 Unauthorized; valid signatures yield HTTP 202 Accepted. Fail: Unsigned payloads are accepted and processed.	Protects internal endpoints from processing unauthorized or spoofed third-party requests.
NFR-001	Non-functional	The ingestion endpoint shall buffer inbound webhooks with 99.999% reliability and throughput up to 2,500 requests/second.	high	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load	Ensures the gateway does not become a bottleneck or single point of failure at peak load.
NFR-002	Non-functional	The gateway shall process and log incoming payloads with an end-to-end ingestion latency of less than 50 ms (95th percentile).	medium	Pass: Load testing shows 95% of incoming HTTP requests receive an acknowledgment response within 50 ms. Fail: Latency p95 exceeds 50 ms under normal operational loads.	Minimizes connection timeout risks for third-party systems dispatching webhooks to the gateway.